Lecture # 8, 9

Friday, 13 March 2026 9:55 am

$$\begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left(\begin{array}{ccc|c} 3 & 4 & 5 & 10 \\ 2 & 3 & 6 & 15 \end{array}\right) \end{array}$$

$$R_1 - R_2 \left(\begin{array}{ccc|c} 1 & 1 & -1 & -5 \\ 2 & 3 & 6 & 15 \end{array}\right)$$

$$R_2 - 2R_1 \left(\begin{array}{ccc|c} 1 & 1 & -1 & -5 \\ 0 & 1 & 8 & 25 \end{array}\right)$$

$$R_1 - R_2 \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left(\begin{array}{ccc|c} 1 & 0 & -9 & -30 \\ 0 & 1 & 8 & 25 \end{array}\right) \end{array}$$

$y + 8z = 25$

$y = 25 - 8z$ $\quad z = t$

$x - 9z = -30$ $\quad x = 9z - 30$

$y = 25 - 8t$

$x = 9t - 30$

$(9t-30,\ 25-8t,\ t)$

Ali and Sara are shopping for chocolate bars. Ali observes, "If I add half my money to yours, it will be enough to buy two chocolate bars." Sara naively asks, "If I add half my money to yours, how many can we buy?" Ali replies, "One chocolate bar." How much money did Ali have?

Ali have $x

SARA Amount $y

$C → 1 chocolate

$\Rightarrow \frac{1}{2}x + y = 2C$

$x + \frac{y}{2} = 1C$

L.H.S $ = R.H.C 2 chocolate bar

$$x + \frac{y}{2} = 1C$$

$$1x + 2y = 4C$$
$$2x + 1y = 2C$$

$$\left(\begin{array}{cc|c} 1 & 2 & 4C \\ 2 & 1 & 2C \end{array}\right) \qquad 2C - 8C$$

$$R_2 - 2R_1 \left(\begin{array}{cc|c} 1 & 2 & 4C \\ 0 & -3 & -6C \end{array}\right)$$

$$-\frac{1}{3}R_2 \left(\begin{array}{cc|c} 1 & 2 & 4C \\ 0 & 1 & 2C \end{array}\right)$$

$$R_1 - 2R_2 \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 2C \end{array}\right) \qquad 1 \cdot C = 5 \quad \$10$$

$$y = 10$$

$$x = \$0 \; ; \qquad y = 2C$$

$$0 + \frac{1}{2}y = 2(5)$$

$$\$10$$

$$\frac{1}{2}x + y = 2C$$

$$2C = 2C$$

Three Alto, two Suzuki, and four City can be rented for \$106 per day. At the same rates two Alto, four Suzuki, and three City cost \$I07 per day, whereas four Alto, three Suzuki, and two City cost \$102 per day. Find the rental rates for all three kinds of cars?

Alto = \$x
Suzuki = \$y
City = \$z

$$3x + 2y + 4z = 106$$
$$2x + 4y + 3z = 107$$
$$4x + 3y + 2z = 102$$

$4x + 3y + 2z = 102$

$x = 10 \quad y = 12, \quad z = 13$

$x \rightarrow 4$

Example 3

A restaurant owner plans to use x tables seating 4, y tables seating 6 and z tables seating 8, for a total 20 tables. When fully occupied, the tables seat 108 customers. If only half of the x tables, half of the y tables and one-fourth of the z tables are used, each fully occupied, then 46 customers will be seated. Find x, y, and z.

$$\boxed{x + y + z = 20}$$

$$4x + 6y + 8z = 108$$

$$\left(\frac{1}{2}x + \frac{1}{2}y + \frac{1}{4}z = 46\right) \times 4$$

$$2x + 2y + z = 184$$

① No. of Tables $\quad x + y + z = 20$

② No. of People when all Tables are fully occupied

$$4x + 6y + 8z = 108 \quad \div 2$$

$$2x + 3y + 4z = 54$$

③ $\left(\frac{1}{2}x + \frac{1}{2}y + \frac{1}{4}z = 46\right) \times 4$

$$2x + 2y + z = 184$$

$$x + y + z = 20$$

$$2x + 3y + 4z = 54$$

$$2x + 2y + 1z = 184$$

Q1. Students are buying books for the new semester. Asma buys the linear algebra book and the differential equation book for $178. Aiman, who is buying books for herself and her friend, spends $319 on two linear algebra books, one differential equation book, and one educational psychology book. Sara buys the educational psychology book and the differential equation book for $147 in total. How much does each book cost?